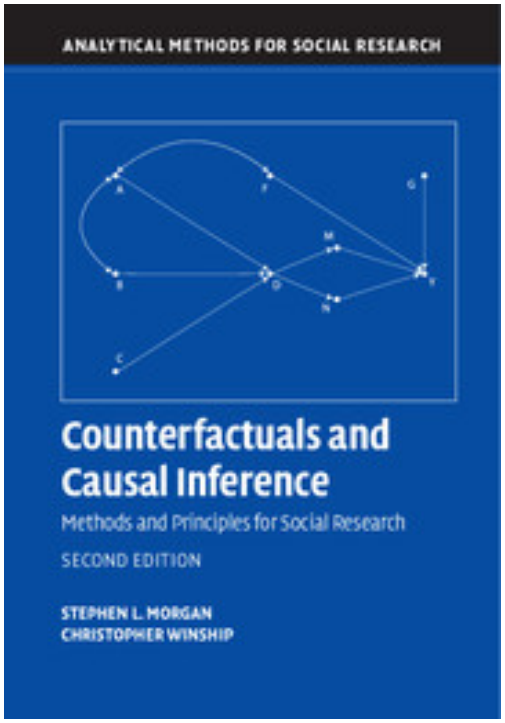
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 1

Outline

1. The Potential Outcome Model of Causal Inference
2. Causal Analysis and Observational Social Science
3. Examples Used Throughout the Book
4. Observational Data and Random-Sample Surveys
5. Causal Graphs as a (Partial) Introduction to the Remainder of the Book

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Potential outcomes

- The potential outcomes of each individual i are ${y_i}^1$ and ${y_i}^0$, where the superscript 1 signifies the treatment state and the superscript 0 signifies the control state.
- Because both ${y_i}^1$and ${y_i}^0$ exist in theory for each individual, an individual-level causal effect can be defined as some contrast between ${y_i}^1$ and ${y_i}^0$, usually the simple difference ${y_i}^1 - {y_i}^0$.
- Because it is impossible to observe both ${y_i}^1$ and ${y_i}^0$for any individual, causal effects cannot be observed or directly calculated at the individual level.

Observed outcomes

- By necessity, a researcher must analyze an observed outcome variable Y that takes on values y_i for each individual i that are equal to ${y_i}^1$ for those in the treatment state and ${y_i}^0$ for those in the control state.
- We usually refer to those in the treatment state as the treatment group and those in the control state as the control group.
- Accordingly, ${y_i}^0$ is an unobservable counterfactual outcome for each individual i in the treatment group, and ${y_i}^1$ is an unobservable counterfactual outcome for each individual i in the control group.

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Experimental language in observational social science

- Early matching designs
- Stouffer and survey research
- Quasi-experiments

"The Age of Regression"

- Blalock and Duncan
- Emergent practice
- Abbott's critique of General Linear Reality
- The call for attention to mechanisms

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Broad examples from the social, demographic, and health sciences

- The Causal Effects of Family Background and Intelligence on Educational Attainment
- The Causal Effects of Educational Attainment and Mental Ability on Earnings
- The Causal Effects of Family Background, Educational Attainment, and Earnings on Political Participation
- The Causal Effects of Family Background and Life Course Events on Fertility Patterns
- The Causal Effects of Socioeconomic Status on Health and Mortality

Narrow and specific examples

- The Causal Effects of Neighborhood of Residence on Educational Performance, Deviance, and Youth Development
- The Causal Effects of Father Absence on Child and Adolescent Development
- The Causal Effect of Catholic Schooling on Learning
- The Causal Effect of School Vouchers on Learning
- The Causal Effect of Charter Schools on Learning
- The Causal Effect of Worker Training on Earnings
- The Causal Effects of Risky Health Behaviors and Peer Relationships on Obesity and Mortality
- The Causal Effect of Voting Technology on Valid Voting and Election Outcomes

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A deliberate focus on:

- Populations and random samples from them
- Identification rather than statistical inference
- Perfect measurement

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Back-door conditioning/adjustment

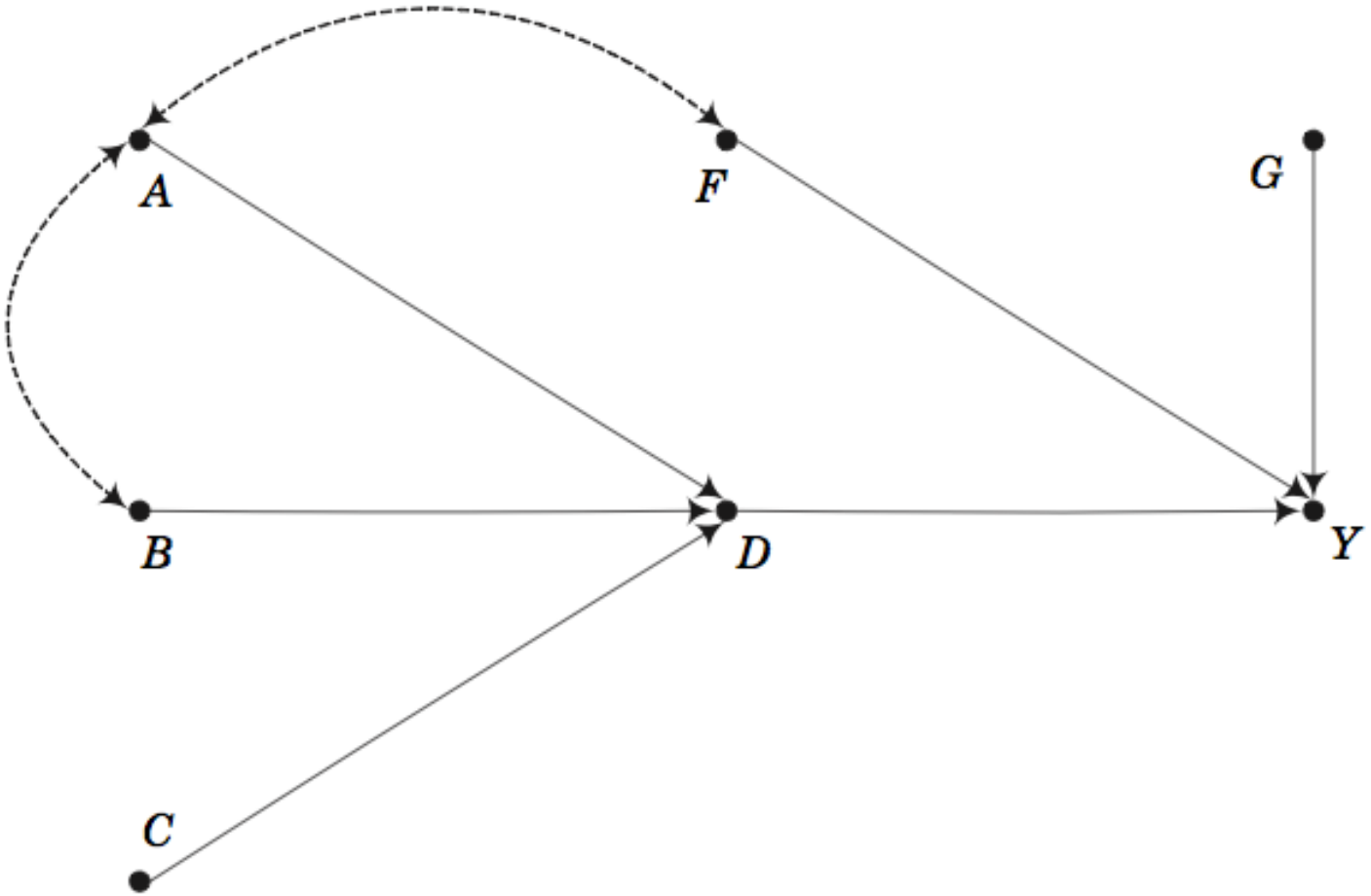

Figure 1.1 A causal graph in which back-door paths from D to Y can be blocked by observable variables and in which C is an instrumental variable for D.

Natural experiments and instrumental variables

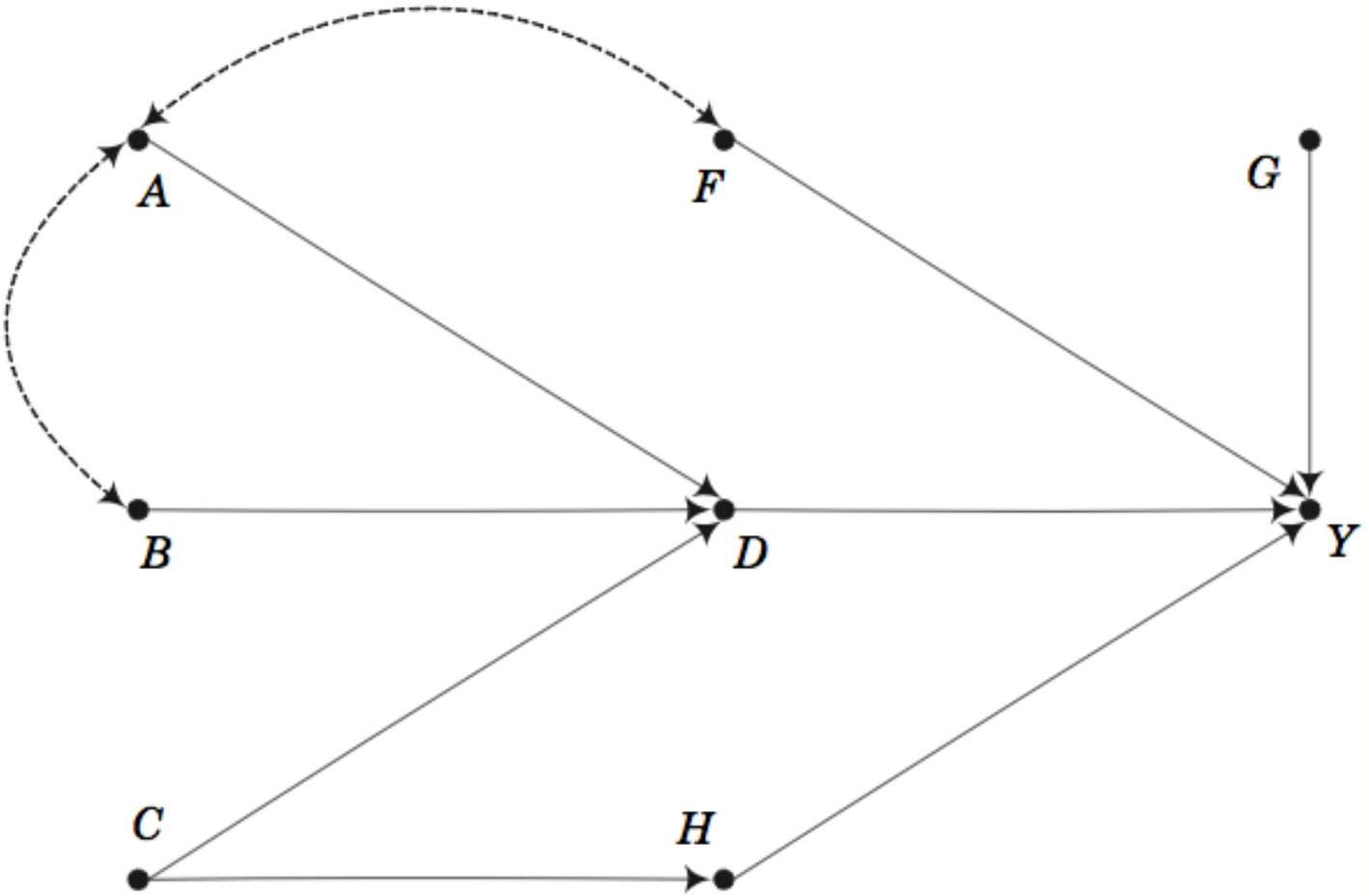

Figure 1.2 A causal graph in which C is no longer an instrumental variable for D.

Mechanisms

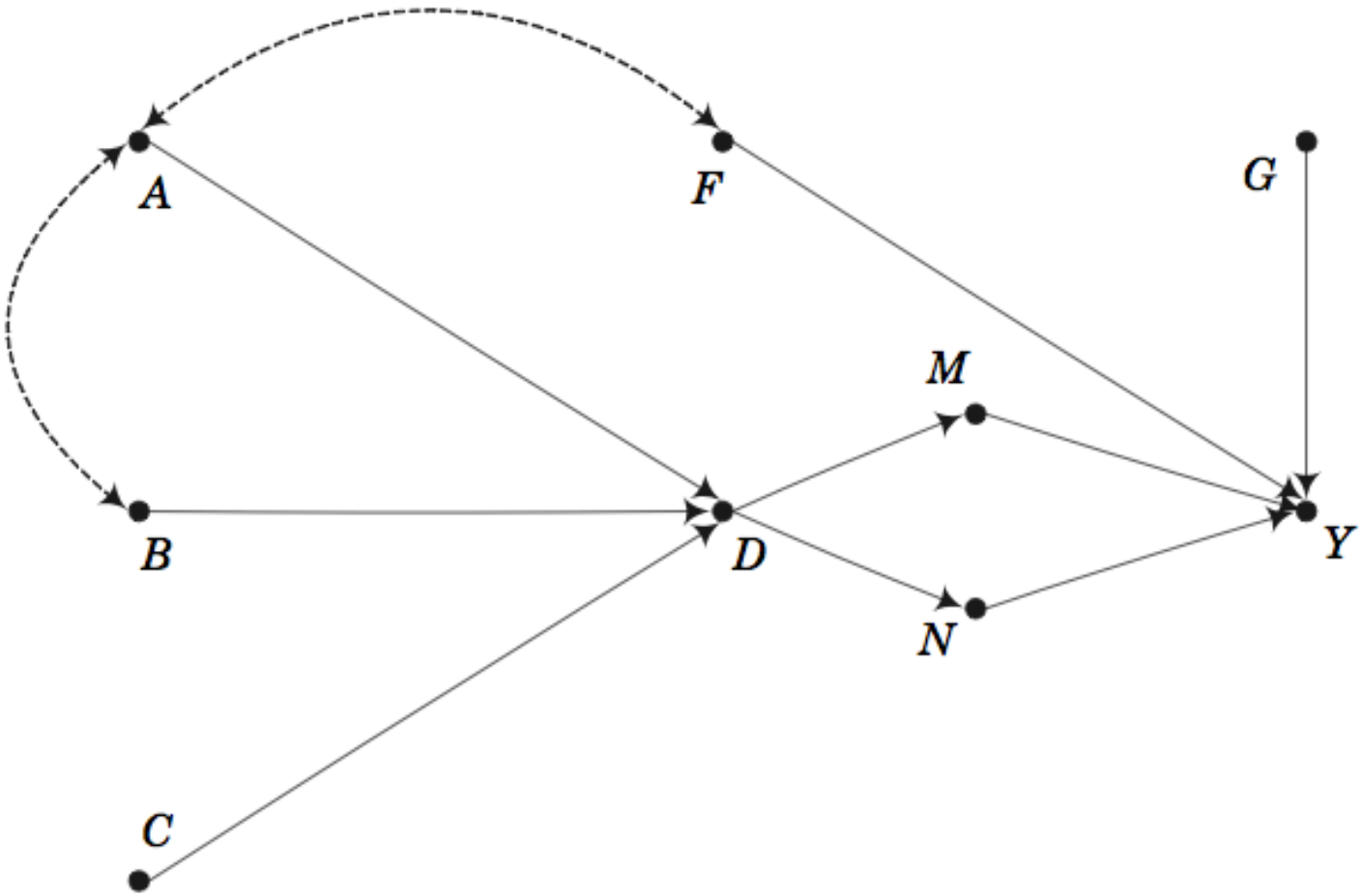

Figure 1.3 A causal diagram in which M and N represent an isolated and exhaustive mechanism for the causal effect of D on Y.